Clio — Build Guide for AI Manthan 2.0, Problem Statement 16

Oct 31, 2026 · @Someone

1. Project overview

Clio (the Greek muse whose name means "to proclaim") is a browser app that watches and listens to a person with a speech impairment, fuses voice + lips + face + hands into ranked intents, suggests replies to the other speaker, and speaks the confirmed message in the patient's own consented voice in under 2 seconds.

One-line pitch: "Your words, your face, your hands, your voice, understood in real time, in English, Hindi or Hinglish."

PS16 requirements mapped to features

Every line of the problem statement has a feature and a proof point judges can see on screen.

PS16 requirement	Clio feature	What judges see
Analyse patient's voice	Scribe v2 Realtime stream on patient mic, keyterm biasing with the patient's personal phrase bank	Live partial transcript with per-word confidence colouring
Gestures, facial expressions, lip movements	MediaPipe Hand, Face and Gesture landmarkers in the browser at 30 fps	Landmark overlay on video, chips like "thumbs up", "pointing", "brow raise", "mouth: /pa/"
Fuse to infer intended message	Fusion engine: LLM ranks top-3 intents from transcript n-best + visual cues + conversation context	Intent cards with confidence bars and "why" badges (voice / lips / gesture)
Listen to the other person	Second Scribe stream on partner mic, speaker-tagged	Chat-style conversation timeline

PS16 requirement	Clio feature	What judges see
Suggest replies	Reply generator conditioned on partner's last turn + patient profile	3 reply chips + 4 quick-phrases, one tap or one gesture to choose
Speak confirmed message, natural personalised voice	ElevenLabs Instant Voice Clone or Voice Design, streamed with Flash v2.5 (fast) or v3/v4 (expressive)	Waveform plays in the patient's voice, with emotion tag (happy / urgent)
ElevenLabs mandatory for STT and TTS	All recognition and speech go through ElevenLabs; no fallback ASR/TTS in the main path	ElevenLabs badge in the latency HUD
English, Hindi, code-mixed	Auto language detection, Hinglish-aware LLM prompts, Devanagari + Roman display toggle	Demo turn in Hinglish
Enforce consent for voice cloning	Consent wizard: recorded spoken consent, signed record, revocation, clone locked until consent exists	Consent ledger page; clone button disabled without consent

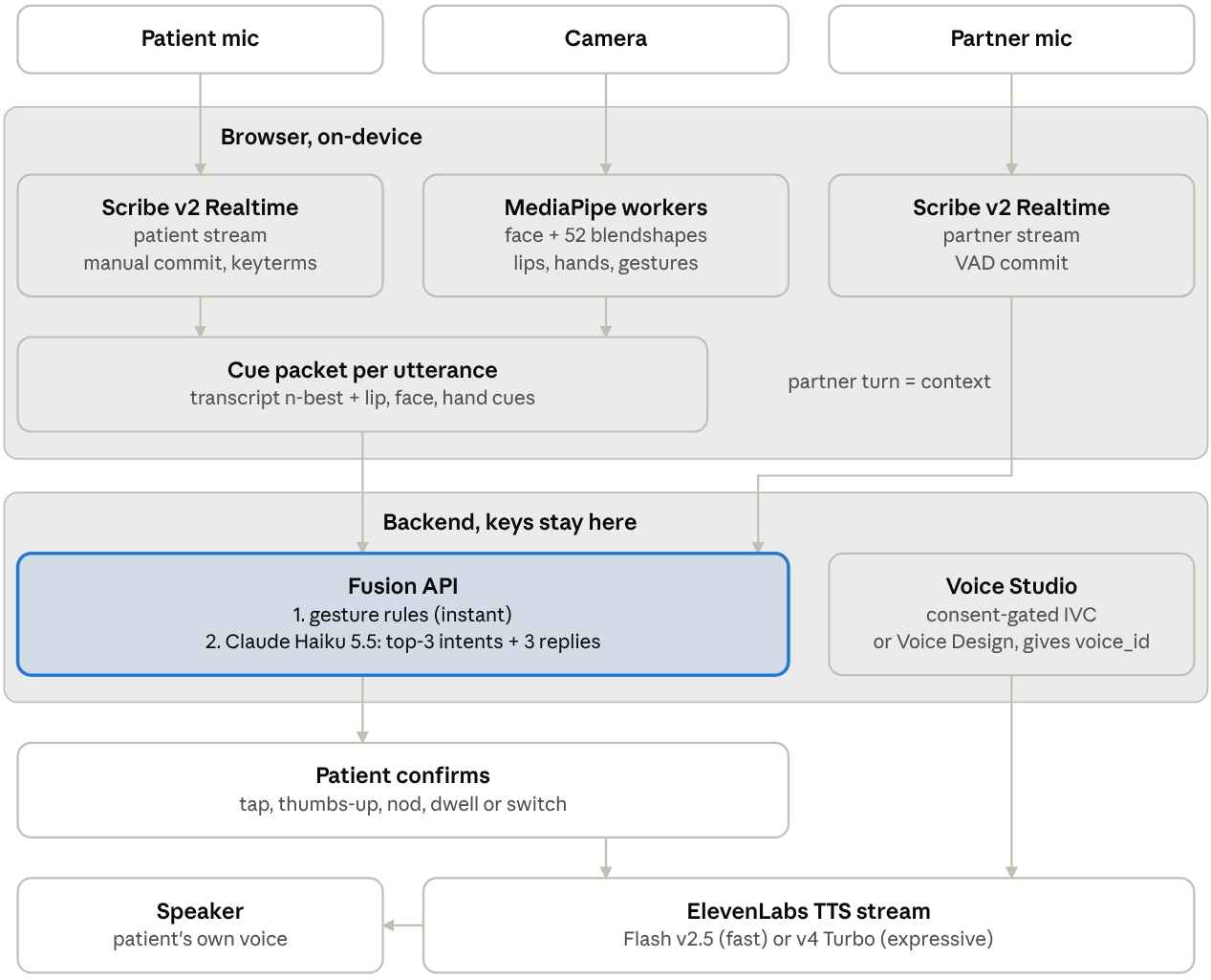
Evaluation checklist (from the hints)

- [ ] WER on atypical speech, with and without gesture/lip fusion
- [ ] Intent accuracy (top-1 and top-3)
- [ ] End-to-end latency under 2 s (end of patient utterance to first audio)
- [ ] Confirmation effort (taps or gestures per confirmed message, time to confirm)
- [ ] Data from TORGO / UA-Speech, or simulated and clearly labelled
- [ ] Only consented voices

2. System architecture

Three live inputs are processed in the browser, fused once on the backend, confirmed by the patient, and spoken by ElevenLabs within a 2-second budget.

Voice, lips, face and hands meet in one fusion step before speech



Clio architecture · 3 inputs, 1 fusion step, 1 voice output

The partner's speech never goes to TTS; it only gives the fusion step context for ranking intents and writing replies.

Latency budget (targets to measure, not measured yet)

Stage	Target	How to hit it
Scribe commit to final text	150–300 ms	Keep the socket open; commit on gesture/button
Cue packet build	under 30 ms	Vision already running in workers; just summarise the last window

Stage	Target	How to hit it
Gesture rules	under 5 ms	Pure lookup
Claude fusion + replies	400–800 ms	Haiku, short JSON schema, prompt caching, start on partials
Patient confirmation	user time, not counted	One action; pre-highlight the top candidate
TTS first audio byte	150–300 ms	Pre-opened socket, Flash v2.5, play the first chunk immediately
Total (commit → first audio, excluding confirm)	**about 0.8–1.4 s**	Under the 2 s PS target with headroom

Speed tricks: run fusion speculatively on partial transcripts and refresh when the commit lands; cache TTS audio for quick phrases; warm both WebSockets at page load.

3. Tech stack and ElevenLabs integration

Use Next.js on the front end, FastAPI on the back end, MediaPipe in the browser, Claude for fusion, and ElevenLabs for every word heard and spoken.

Layer	Choice	Why
Frontend	Next.js 15 (App Router) + TypeScript + Tailwind CSS + shadcn/ui + Framer Motion	Fast to build, polished components, smooth motion
Vision	`@mediapipe/tasks-vision`: FaceLandmarker (478 points + 52 blendshapes), HandLandmarker (21 points/hand), GestureRecognizer	Runs on-device via WebGPU/WASM, no video leaves the browser
Patient + partner STT	ElevenLabs Scribe v2 Realtime over WebSocket	Mandatory; low latency; Hindi + auto language detection; keyterms
Batch STT (eval)	ElevenLabs `scribe_v2` batch API	WER benchmarking on TORGO / UA-Speech

Layer	Choice	Why
Fusion + replies	Claude API, `claude-haiku-5-5` for live ranking, `claude-sonnet-5-5` for offline eval/labelling	Fast structured JSON output, strong Hinglish handling
TTS (fast path)	`eleven_flash_v2_5` over the Text to Speech WebSocket	Lowest latency; Hindi supported
TTS (expressive path)	`eleven_v4_turbo` (or `eleven_v3`) over the Text to Dialogue WebSocket, with audio tags like `[warm]`, `[urgent]`	Emotional delivery matching the patient's face
Voice creation	Instant Voice Cloning (consented samples) or Voice Design (`eleven_multilingual_ttv_v2`) when no recordings exist	Personalised voice, consent-gated
Backend	Python 3.12, FastAPI, `websockets`, `elevenlabs` SDK, `anthropic` SDK	Orchestration, token minting, consent ledger
Storage	SQLite (hackathon) via SQLModel; Postgres-ready	Profiles, consent records, eval logs
Eval	`jiwer` (WER/CER), pandas, matplotlib	Metrics required by the PS

ElevenLabs facts to build against (verified Oct 2026)

- **Scribe v2 Realtime**: model id `scribe_v2_realtime`, endpoint `wss://api.elevenlabs.io/v1/speech-to-text/realtime`. ElevenLabs states about 150 ms latency and 90+ languages including Hindi (ElevenLabs).
- **Commit strategies**: `manual`, `vad`, or `turn_prediction`. Use `manual` for the patient (dysarthric speech has long pauses that VAD cuts too early) and `vad` / `turn_prediction` for the partner (API reference).
- **Keyterms**: up to 50 keyterms on realtime, passed as repeated `keyterms=` query params. Load the patient's personal phrase bank (names, medicines, carers) here (Keyterm guide).
- **Client-side auth**: the realtime API accepts a single-use `token`; mint it on the backend so the API key never reaches the browser.

- **Flash v2.5**: about 75 ms model inference, streamed over `wss://api.elevenlabs.io/v1/text-to-speech/{voice_id}/stream-input` (Latency guide).
- **Eleven v4 / v4 Turbo**: released 28 Sep 2026; v4 Turbo is the real-time variant at about 100 ms median inference, streamed only through the Text to Dialogue WebSocket (changelog). That socket accepts model ids starting with `eleven_v3` or `eleven_v4`, and Turbo takes one voice per connection (guide).
- **Voice Design**: call `text_to_voice.design` with a 20–1000 character description, get 3 previews, then `text_to_voice.create` with the chosen `generated_voice_id` (quickstart).
- **Instant Voice Cloning**: `voices.ivc.create` from short samples (under two minutes) (voice types).

The PS names "Flash or v3". Ship Flash v2.5 as default and offer v4 Turbo (or v3) as the "expressive" toggle; say this explicitly in the pitch.

Consent enforcement (non-negotiable)

1. Consent wizard: the voice owner (patient, or a family member whose old recordings are used) reads a consent sentence aloud in their chosen language; Scribe transcribes it and the backend checks it matches.
2. Store a consent record: owner name, relationship, purpose, timestamp, audio hash, scope ("Clio use only"), expiry.
3. Backend refuses `ivc.create` unless a valid consent id is attached; the UI disables "Clone voice" until then.
4. One-tap revoke: deletes the ElevenLabs voice via API and marks the record revoked.
5. For people who have already lost their voice and have no recordings, Voice Design creates a new voice from age, gender, accent and warmth; no one else's voice is copied.

Fusion design

The LLM never sees raw video. The browser sends a compact **cue packet** every committed utterance:

```
{
  "asr": {"text": "paani ch...", "words": [{"w": "paani", "conf": 0.81}],
"lang": "hi"},
  "lips": {"jawOpen": 0.42, "mouthPucker": 0.10, "visemes": ["p", "aa",
"n"]},
  "face": {"browInnerUp": 0.6, "smile": 0.05, "affect": "distress"},
  "hands": {"gesture": "Pointing_Up", "target": "left", "held_ms": 900},
  "context": {"partner_last": "Kuch chahiye aapko?", "time": "14:05",
"location": "ward"}
}
```

Claude returns strict JSON: top-3 intents with confidence, supporting modalities, a reconstructed sentence in the patient's language, and 3 reply suggestions. Show the "why" chips so the fusion is visible to judges.

Add a cheap deterministic layer before the LLM: a gesture map (thumbs up = yes, open palm = stop, pointing up = need help, hand to mouth = thirsty) that can confirm or veto LLM candidates. This keeps the system working if the network is slow.

4. UI/UX design spec

The UI must feel calm, warm and medical-grade, with huge touch targets and one-gesture confirmation, because the user may have limited motor control.

Design system

Token	Value	Use
Background	Deep ink `#0B1020` (dark) / warm paper `#FAF7F2` (light)	Low glare for long sessions
Primary	Saffron-to-coral gradient `#FF9933` → `#FF5E5B`	Patient's voice, "Speak" button
Secondary	Teal `#14B8A6`	Partner's speech
Accent	Violet `#8B5CF6`	AI suggestions, fusion chips
Status	Green `#22C55E` / Amber `#F59E0B` / Red `#EF4444`	Confidence, latency, consent
Display font	Plus Jakarta Sans 600–800	Headings, big replies

Token	Value	Use
Hindi font	Noto Sans Devanagari / Mukta	Devanagari text, same weight scale
Body size	18 px minimum, 32–40 px for reply chips	Readable at arm's length
Radius	20 px cards, full-round chips	Soft, friendly
Motion	Framer Motion springs, 200–300 ms; respect `prefers-reduced-motion`	Never distracting

Glassmorphism cards (backdrop blur 16 px, 8% white fill) on a slow animated mesh gradient give the "wow" without hurting readability.

Screens

1. **Welcome / onboarding**: language picker (English, हिन्दी, Hinglish), condition profile (ALS, CP, Parkinson's, stroke, laryngectomy), camera and mic permission with a friendly explainer.
2. **Voice studio**: consent wizard, sample recorder with live level meter, or Voice Design form ("warm, 60-year-old Indian man, soft voice"); 3 preview cards to audition; consent ledger with revoke.
3. **Calibration (2 min)**: patient says 10 personal phrases and shows 6 gestures; builds keyterms and a per-user gesture map. Progress ring animation.
4. **Live conversation (hero screen)**:
   - Left: camera tile with neon landmark overlay (face mesh, lip contour highlighted, hand skeleton), toggleable.
   - Centre: conversation timeline. Partner bubbles (teal) and patient bubbles (saffron) with live partial text shimmering as Scribe streams.
   - Right: **Intent panel**: top-3 intent cards with confidence bars and modality chips (Voice 62% · Lips · Gesture ✓). Below: 3 suggested replies + quick-phrase grid (Yes, No, Wait, Pain, Water, Washroom, Thank you).
   - Bottom: giant **Speak** button with live waveform; emotion selector (calm / happy / urgent) auto-set from facial affect.
   - Top bar: latency HUD (STT ms · Fusion ms · TTS first-byte ms · Total), green when under 2 s.
5. **Emergency mode**: a long-held open palm or a double brow-raise flashes a red full-screen card and speaks "I need help now" at max volume.

6. **Insights dashboard**: WER with vs without fusion, intent accuracy, latency histogram, taps-per-message, all with "SIMULATED" badges where relevant.
7. **Caregiver/partner view** (stretch): a phone-sized page showing large captions of what the patient says.

Confirmation without effort

The patient confirms the top intent with any one of: a tap, thumbs-up, a nod (head pitch from face landmarks), a 1.5 s dwell gaze on a card, or a single key/switch press. Thumbs-down or a head shake cycles to the next candidate. Track how many actions each confirmed message takes; this is the PS's "effort" metric.

Accessibility checklist

- [ ] WCAG 2.2 AA contrast; all actions reachable by one switch (Space/Enter scanning mode)
- [ ] Touch targets 64 px or larger
- [ ] Captions for every audio output; screen-reader labels on all controls
- [ ] Devanagari/Roman toggle for every transcript
- [ ] No time-outs that punish slow responses

Use the frontend-design and shadcn/ui component patterns, and ask Claude Code to produce the landing page first so the look is locked before features are wired.

5. Data, evaluation and metrics

Measure Scribe's baseline on real dysarthric English, then prove that fusion lowers errors on a labelled simulated multimodal set covering English, Hindi and Hinglish.

Datasets

Dataset	What it gives you	How to use it	Label
TORGO	Dysarthric English speech (CP and ALS speakers) plus controls	Baseline Scribe WER by severity	Real

Dataset	What it gives you	How to use it	Label
UA-Speech	Dysarthric English isolated words (CP); access by request	Word-level WER; keyterm before/after	Real
Clio-Sim (you build it)	Team members act out 50–100 scripted turns on camera: slurred/slow speech, gestures, expressions, in EN/HI/Hinglish	Fusion evaluation, intent accuracy, latency	**SIMULATED**, labelled in every chart and file name

Atypical-speech simulation recipe: speak slowly with reduced articulation, or post-process clean recordings (time-stretch 0.7×, low-pass 3 kHz, add breathiness/jitter). Store a `sim_method` field on every clip. Never present simulated numbers as clinical results.

Metrics

Metric	Definition	Target
WER (voice only)	`jiwer.wer(ref, scribe_text)` on patient turns	Baseline, report by severity
WER (fused)	WER of the LLM-reconstructed sentence vs reference	Lower than voice only; report the relative drop
Intent accuracy	Top-1 and top-3 match against a labelled intent per turn	Top-3 above 85% on Sim set
End-to-end latency	Patient commit → first TTS audio byte played	p50 under 1.2 s, p95 under 2 s
Confirmation effort	Actions per confirmed message; seconds to confirm	⩽ 1.3 actions, under 3 s
Failure analysis	Where Scribe fails: phoneme classes, word length, language, severity	A chart of the top failure patterns

Run an ablation table for judges: voice only → +lips → +face → +gesture → +context. That one table proves the multimodal claim.

Log every live turn (`turn_id` , timestamps per stage, cue packet, candidates, chosen, actions) to SQLite so the dashboard and the report come from the same data.

6. Repo structure, API contracts and setup

One monorepo, two apps, one shared schema package, so Claude Code always knows where things go.

```
clio/
├── CLAUDE.md                # project memory for Claude Code (section 7)
├── apps/
|   ├── web/                 # Next.js 15 frontend
|   |   ├── app/(onboarding)/ (studio)/ (live)/ (insights)/
|   |   ├── components/ui/    # shadcn components
|   |   ├── components/live/  # CameraTile, Timeline, IntentPanel,
SpeakButton, LatencyHUD
|   |   ├── lib/vision/       # mediapipe workers: face, hands, gestures,
visemes
|   |   ├── lib/audio/        # mic capture (AudioWorklet, PCM16 16 kHz),
player
|   |   └── lib/elevenlabs/   # scribe realtime client, tts socket client
|   └── api/                  # FastAPI backend
|       ├── main.py
|       ├── routers/ tokens.py fusion.py voices.py consent.py tts.py eval.py
|       ├── services/ fusion_llm.py gesture_rules.py elevenlabs_client.py
|       ├── models/ db.py schemas.py
|       └── tests/
├── packages/schemas/         # shared JSON schemas (cue packet, intents)
├── eval/
|   ├── datasets/ (torgo/, uaspeech/, sim/)
|   ├── run_wer.py  run_ablation.py  latency_report.py
|   └── reports/
└── docker-compose.yml
```

API contracts

Method + path	Purpose	Returns
`POST /tokens/scribe`	Mint single-use realtime STT token	`{token, expires_at}`

Method + path	Purpose	Returns
`POST /fusion/rank`	Cue packet + context → intents + replies	`{intents[3], sentence, replies[3], latency_ms}`
`POST /consent`	Store spoken consent (audio + transcript)	`{consent_id, status}`
`DELETE /consent/{id}`	Revoke; deletes the ElevenLabs voice	`{status: "revoked"}`
`POST /voices/clone`	IVC, requires `consent_id`	`{voice_id}` or 403
`POST /voices/design`	Voice Design previews	`{previews[3]}`
`POST /voices/design/save`	Save chosen preview	`{voice_id}`
`WS /tts/stream`	Proxy text → ElevenLabs TTS socket → audio chunks	binary audio
`POST /turns`	Log a turn with per-stage timings	`{turn_id}`
`GET /eval/summary`	Metrics for the dashboard	JSON

Environment

```
# apps/api/.env
ELEVENLABS_API_KEY=...
ANTHROPIC_API_KEY=...
FUSION_MODEL=claude-haiku-5-5
TTS_FAST_MODEL=eleven_flash_v2_5
TTS_EXPRESSIVE_MODEL=eleven_v4_turbo
DATABASE_URL=sqlite:///clio.db

# apps/web/.env.local
NEXT_PUBLIC_API_URL=http://localhost:8000
```

The ElevenLabs and Anthropic keys live only on the backend; the browser gets short-lived tokens.

7. Claude Code build guide

Put the CLAUDE.md below in the repo root, then paste the phase prompts in order; each phase ends with something demoable, so you are never stuck with half a system.

Setup

```
mkdir clio && cd clio && git init
claude            # start Claude Code in the repo
# inside Claude Code: /init, then replace CLAUDE.md with the version below
```

Tips: use plan mode (Shift+Tab) before each phase, commit after each green phase, and keep one terminal running `pnpm dev` and one running `uvicorn` so Claude Code can test against live servers.

CLAUDE.md (copy as-is)

```
# Clio: multimodal conversation assistant for people with speech impairments
AI Manthan 2.0, Problem Statement 16.

## Non-negotiables
- ALL speech recognition and ALL speech output use ElevenLabs. No other
ASR/TTS in the main path.
- STT: Scribe v2 Realtime (model_id=scribe_v2_realtime,
wss://api.elevenlabs.io/v1/speech-to-text/realtime).
  Patient stream: commit_strategy=manual. Partner stream:
commit_strategy=vad. Pass patient keyterms (max 50).
  Browser connects with a single-use token minted by POST /tokens/scribe.
Never expose API keys to the client.
- TTS fast: eleven_flash_v2_5 via /v1/text-to-speech/{voice_id}/stream-input.
  TTS expressive: eleven_v4_turbo via /v1/text-to-dialogue/stream-input (one
voice per connection).
- Voice cloning ONLY with a valid consent_id. /voices/clone returns 403
without it. Revoke deletes the voice.
- Support English, Hindi (Devanagari) and Hinglish everywhere: prompts, UI
strings, transcripts.
- Any simulated data or metric is labelled SIMULATED in file names, DB rows
and UI.
- Latency budget: patient commit -> first audio < 2000 ms. Log per-stage
timings for every turn.

## Stack
```

```
web: Next.js 15, TypeScript strict, Tailwind, shadcn/ui, Framer Motion,
@mediapipe/tasks-vision, zustand.
api: Python 3.12, FastAPI, SQLModel/SQLite, elevenlabs SDK, anthropic SDK
(FUSION_MODEL env), jiwer.

## Design system
Dark ink #0B1020 / light #FAF7F2. Patient = saffron-coral gradient, partner =
teal #14B8A6, AI = violet #8B5CF6.
Fonts: Plus Jakarta Sans + Noto Sans Devanagari. Body >= 18px, reply chips
32-40px, targets >= 64px.
Glass cards, 20px radius, spring motion 200-300ms, honour prefers-reduced-
motion. WCAG 2.2 AA.

## Conventions
- Shared JSON schemas live in packages/schemas; validate with zod (web) and
pydantic (api).
- Vision runs in Web Workers; never block the main thread. Never upload raw
video.
- Every feature ships with a test (pytest / vitest) and a mock mode
(MOCK_ELEVENLABS=1) for offline demos.
- Commands: `pnpm dev` (web), `uvicorn main:app --reload` (api), `pytest`,
`pnpm test`.
```

Phase prompts (paste one at a time)

1. **Scaffold (1.5 h)**: "Read CLAUDE.md. Scaffold the monorepo exactly as in the structure section: Next.js app with Tailwind, shadcn/ui, Framer Motion, the design tokens as CSS variables with dark/light themes; FastAPI app with health route, SQLModel DB, CORS. Add docker-compose. Make both run and show a styled placeholder home page."
2. **Landing + design polish (1.5 h)**: "Build a stunning landing page and app shell: animated mesh gradient background, glass cards, hero with the tagline 'Your words, your face, your hands, your voice', language switcher EN/हिन्दी/Hinglish, sidebar nav to Live, Voice Studio, Calibration, Insights. Fully responsive and keyboard accessible."
3. **Scribe realtime (3 h)**: "Implement POST /tokens/scribe that mints a single-use ElevenLabs realtime STT token. In web, build an AudioWorklet mic capture at 16 kHz PCM16 and a ScribeRealtimeClient that streams to scribe_v2_realtime, handles partial and committed transcripts, auto language detection and keyterms. Support two instances: patient (manual commit via a 'Done' gesture/button) and partner (vad). Render a live timeline with shimmering partials."

4. **Vision (3 h)**: "In Web Workers, run MediaPipe FaceLandmarker (with blendshapes), HandLandmarker and GestureRecognizer at 30 fps. Draw a neon overlay with the lip contour highlighted. Derive a cue packet per utterance: mouth blendshape summary and coarse visemes, facial affect, head nod/shake, gesture with hold duration. Match packages/schemas/cue_packet.json."
5. **Fusion (3 h)**: "Implement POST /fusion/rank. First apply services/gesture_rules.py (deterministic map: thumbs up=yes, open palm=stop, pointing up=help, hand to mouth=thirsty). Then call the Anthropic API with FUSION_MODEL and a system prompt that returns strict JSON: top-3 intents with confidence and supporting modalities, reconstructed sentence in the patient's language/script, and 3 replies conditioned on the partner's last turn. Add a 900 ms timeout fallback to rules + quick phrases. Build the IntentPanel with confidence bars and 'why' chips."
6. **Voice studio + consent (2.5 h)**: "Build the consent wizard (spoken consent recorded, transcribed with Scribe, matched to the consent sentence), consent ledger with revoke, IVC upload gated by consent_id (403 otherwise), and Voice Design with 3 preview cards and save. Revoke must delete the ElevenLabs voice."
7. **TTS + confirmation (2.5 h)**: "Implement WS /tts/stream proxying to Flash v2.5 by default and eleven_v4_turbo when 'expressive' is on, adding an audio tag from facial affect. Stream audio to the browser and start playback on first chunk. Confirmation: tap, thumbs-up, nod, 1.5 s dwell, or Space key; thumbs-down/shake cycles candidates. Count actions per message. Add the LatencyHUD with per-stage ms."
8. **Emergency + quick phrases (1 h)**: "Add emergency mode (long open palm or double brow raise) and an editable quick-phrase grid in EN/HI with pre-generated TTS cached for instant playback."
9. **Evaluation (3 h)**: "Write eval/run_wer.py using scribe_v2 batch on TORGO/UA-Speech samples with jiwer; run_ablation.py over the SIMULATED set (voice, +lips, +face, +gesture, +context); latency_report.py from logged turns. Build the Insights page with charts and SIMULATED badges."
10. **Hardening (2 h)**: "Add MOCK_ELEVENLABS mode with recorded fixtures, error toasts, reconnect logic for both sockets, loading skeletons, and a /demo route that replays a scripted conversation. Run all tests and fix failures."

About 23 hours of build; add buffer for sleep and the pitch. Two people can run phases 3–4 and 5–6 in parallel using git worktrees.

Useful follow-up prompts

- "Review the live screen as a designer: list 10 improvements for delight and accessibility, then implement the top 5."
- "Profile one turn end to end and cut latency: show where the time goes and fix the biggest stage."
- "Write the README with architecture diagram, setup, metrics table and consent policy."

8. Demo script, pitch and risks

Win the room with a 3-minute live demo that shows fusion fixing a failed transcript, in Hinglish, in the patient's own voice, with the latency HUD green.

3-minute demo

1. **Hook (20 s)**: "Ramesh had a stroke. He still has things to say. Standard speech apps hear only noise."
2. **Voice only (30 s)**: a teammate speaks slurred "paani chahiye"; Scribe partial shows a wrong guess. Point to the low-confidence red words.
3. **Fusion (40 s)**: he repeats it while touching his mouth; the intent panel jumps to "I'm thirsty, I need water" with Voice · Lips · Gesture chips. Thumbs-up confirms.
4. **Voice (20 s)**: it speaks in his designed or consented cloned voice; HUD shows total under 2 s.
5. **Conversation (40 s)**: partner asks "Thanda ya normal?"; three replies appear; a nod picks "Normal, please". Show emergency mode with an open palm.
6. **Proof (30 s)**: Insights page: WER voice-only vs fused, intent accuracy, latency p50/p95, actions per message, SIMULATED labels visible. Close on the consent ledger and revoke button.

Pitch points

- Multimodal by design: the ablation table proves each modality adds value.
- Mandatory ElevenLabs used end to end: Scribe v2 Realtime for both speakers, Flash v2.5 and v4 Turbo for speech, IVC and Voice Design for identity.
- Indian-first: Hindi and Hinglish, Devanagari display, cultural quick phrases.

- Ethical: consent enforced in code, revocable, no raw video leaves the device, simulated data honestly labelled.
- Scalable: stateless API, token-based browser streaming, swap SQLite for Postgres, caregiver view for remote use.

Risks and fallbacks

Risk	Fallback
Venue Wi-Fi slow or down	Phone hotspot; `MOCK_ELEVENLABS=1` replay mode with recorded fixtures; cached quick-phrase audio
LLM call over 900 ms	Rules-only intent + quick phrases; show "fast mode" badge
Scribe cuts dysarthric speech early	Patient stream uses manual commit, triggered by gesture or button
MediaPipe slow on old laptop	Drop to 15 fps, disable face mesh drawing, keep blendshapes
ElevenLabs credits run out	Set budget alerts; pre-generate demo audio; use Flash (cheaper) by default
v4 Turbo socket issues	Fall back to Flash v2.5 on the Text to Speech socket automatically
Judges question data validity	Show TORGO/UA-Speech baseline as real, Sim set clearly labelled, with the simulation method

Sources

- ElevenLabs Realtime Speech to Text
- Realtime STT API reference
- Keyterm prompting guide
- TTS latency optimization
- ElevenLabs changelog, 28 Sep 2026 (Eleven v4)
- Stream dialogue in real time (Text to Dialogue WebSocket)
- Voice Design quickstart
- Voice types and cloning